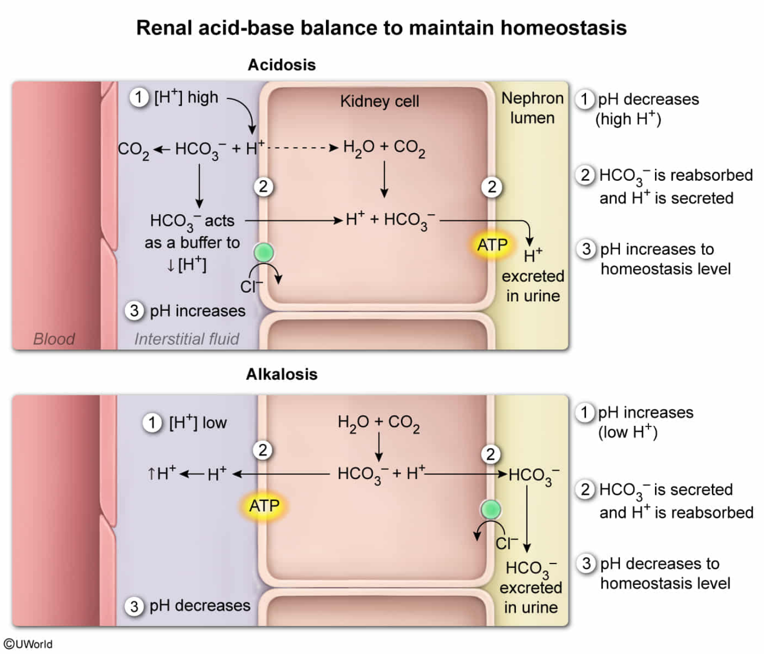
The consumption of excessive amounts of alcohol can lead to very high levels of acidity in the blood. Which of the following pairs of renal responses would most likely take place following excessive alcohol consumption?

I. secretion of H^+
II. secretion of HCO_3^-
III. reabsorption of H^+
IV. reabsorption of HCO_3^-

- A. I and II only [9%]
- B. II and III only [27%]
- C. III and IV only [5%]
- ✔ D. I and IV only [57%]

Correct
58% answered correctly

Explanation:



Blood pH (ie, H^+ concentration) must be maintained within a narrow range (7.38-7.42) by various mechanisms and is impacted by CO_2 levels. Most CO_2 transported in blood combines with water to form carbonic acid (H_2CO_3), which rapidly dissociates into HCO_3^- and H^+. This reaction (ie, **bicarbonate buffer system**) is in equilibrium and aids in maintaining blood pH:

$$CO_2 + H_2O \leftrightharpoons H_2CO_3 \leftrightharpoons HCO_3^- + H^+$$

Therefore, higher blood CO_2 concentrations will *increase* blood acidity (ie, *decrease* pH) due to *increased* H^+ production.

The kidney is involved in the homeostatic regulation of blood pH. In the nephrons, blood is initially filtered through glomerular capillaries. Further reabsorption and secretion of solutes from this filtrate to maintain acid-base balance occurs along the following segments of the nephron:

1. **Proximal tubule**: Bicarbonate (HCO_3^- ions, which can regulate blood pH, are reabsorbed along with other solutes and nutrients. Hydrogen ions (H^+) and waste products are secreted into the filtrate.
2. Loop of Henle:
    a. The *descending* limb extending into the salty medulla reabsorbs water (H_2O) via osmosis.
    b. The *ascending limb,* which transports filtrate out of the medulla and back to the renal cortex, reabsorbs salts and potassium ions.
3. **Distal tubule**: Salts, H_2O, and HCO_3^- are reabsorbed while potassium and H^+ ions are secreted.
4. **Collecting duct**: H_2O and salts continue to be reabsorbed, while potassium ions are secreted. In addition, to regulate blood pH, the reabsorption and secretion of HCO_3^- and H^+

are regulated.

In this question, consuming excess alcohol leads to highly acidic blood (ie, excess H^+ or acidosis). To *lower* acidity, *more* H^+ ions should be *removed* from blood. Accordingly, nephrons would likely respond by **secreting H^+** in the proximal and distal tubules and the collecting duct to increase blood pH, allowing H^+ ions to enter the urine **(Number I)**. Additionally, **HCO_3^-** ions would be **reabsorbed** in the proximal and distal tubules and in the collecting duct and transferred back into the bloodstream **(Number IV)**. This would lead to the accumulation of HCO_3^- in the blood, which acts as a buffer, binding H^+ (ie, removes additional free H^+ ions and lowers blood acidity).

(Numbers II and III) Alkalosis, or a high blood pH, would result in HCO_3^- *secretion* and H^+ *reabsorption*. However, this question describes alcohol-induced acidosis (*not* alkalosis); as a result, HCO_3^- ions are reabsorbed (*not* secreted) and H^+ ions are secreted (*not* reabsorbed).

Educational objective:

When blood pH deviates from homeostatic levels, nephrons of the kidney can maintain blood pH balance via the regulated secretion and reabsorption of H^+ and HCO_3^-.

Time spent: 0 sec	**QID:** 403391
Subject: Biology	**System:** Digestion and Excretion